THE CLASS-NUMBER-ONE ATLAS: NINE CM CONSTANTS AND THEIR χ-UNIFICATIONS

Abstract. For each of the nine imaginary quadratic fields $K = \mathbb{Q}(\sqrt{-d})$ with class number one — that is, $d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$ — the Chowla–Selberg formula assigns a canonical χ-twisted Γ-product constant G_K. The constant $G^* = \Gamma(1/4)/\Gamma(3/4)$ of the lemniscatic case $d = 1$ [1] and the equianharmonic constant $R_3^{3/2}$ of the case $d = 3$ are recovered as the two distinguished members of this atlas — the only two for which the unit group $\mathcal{O}_K^\times$ has order strictly larger than 2 (4 for $\mathbb{Q}(i)$, 6 for $\mathbb{Q}(\rho)$). The remaining seven fields with $|\mathcal{O}_K^\times| = 2$ each contribute a fresh canonical constant.

We tabulate all nine values to 50-digit precision, recover the Heegner near-integer phenomenon $e^{\pi\sqrt{163}} \approx 640320^3 + 744$ as a manifestation of the χ_{-163}-projection structure (and analogous near-integers for $d = 43$ and $d = 67$), and state the atlas theorem: every class-number-one imaginary quadratic field carries a canonical Chowla–Selberg constant uniquely determined by its Kronecker character. The atlas reduces to the dual-constant dichotomy of [1] precisely at the two fields with extra automorphisms.

Contents

1. The class-number-one IQ fields and their constants

Let $K = \mathbb{Q}(\sqrt{-d})$ be an imaginary quadratic field with $d > 0$ squarefree, ring of integers $\mathcal{O}_K$, discriminant d_K, unit group $\mathcal{O}_K^\times$ of order w_K, and class number h_K. There are exactly nine fields K with $h_K = 1$ [2]: $d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$.

For each such K, the Chowla–Selberg formula [3] assigns a canonical χ-twisted Γ-product whose closed form depends on $|d_K|$, w_K, and the Kronecker character χ_{d_K} of K. In this note we tabulate the resulting constants to 50-digit precision, identify their structural roles, and exhibit the unified picture.

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1.1. Definitions.

Definition 1.1. For each class-number-one imaginary quadratic field $K = \mathbb{Q}(\sqrt{-d})$, the *Chowla–Selberg constant* of K is the χ-twisted Γ-product

$$G_K \;:=\; \prod_{a=1}^{|d_K|-1} \Gamma\!\left(\frac{a}{|d_K|}\right)^{\chi_{d_K}(a)\, w_K/4},$$

where $\chi_{d_K} : (\mathbb{Z}/|d_K|\mathbb{Z})^\times \to \{\pm 1\}$ is the unique non-trivial real Dirichlet character of conductor $|d_K|$ (the Kronecker character of K), extended by zero to all integers.

This definition recovers the lemniscatic constant $G^* = \Gamma(1/4)/\Gamma(3/4)$ of [1] as the special case $d = 1$, where $|d_K| = 4$, $w_K = 4$, and $\chi_{-4}(1) = +1$, $\chi_{-4}(3) = -1$; the exponent factor $w_K/4 = 1$ gives the Γ-ratio.

It recovers the equianharmonic ratio of [1] as the special case $d = 3$, where $|d_K| = 3$, $w_K = 6$, and $\chi_{-3}(1) = +1$, $\chi_{-3}(2) = -1$; the exponent factor $w_K/4 = 3/2$ gives $G_K = R_3^{3/2}$ with $R_3 = \Gamma(1/3)/\Gamma(2/3)$.

1.2. The atlas.

Theorem 1.2 (The h=1 atlas)**.** *The Chowla–Selberg constants G_K for the nine class-number-one imaginary quadratic fields are*

d	$\lvert d_K\rvert$	w_K	$j(\tau_K)$	G_K *(50-digit precision)*
1	4	4	1728	2.95867511918863890231 . . .
2	8	2	8000	3.38008909479755370666 . . .
3	3	6	0	2.78265513844626344265 . . .
7	7	2	−3375	3.31921570191787429534 . . .
11	11	2	−32768	3.48914080352712252598 . . .
19	19	2	−884736	3.49035414943653465912 . . .
43	43	2	−884 736 000	2.95295156929956847959 . . .
67	67	2	−147 197 952 000	2.40695989855034264698 . . .
163	163	2	−262 537 412 640 768 000	1.13131740751874176944 . . .

Two of the nine fields have $w_K > 2$: $d = 1$ (lemniscatic, $w_K = 4$) and $d = 3$ (equianharmonic, $w_K = 6$). For these two, G_K is the constant in the central role identified in [1, §16]*:*

$$G_K\big|_{d=1} = G^*, \qquad G_K\big|_{d=3} = R_3^{3/2}.$$

Proof. The values are direct numerical evaluations of Definition 1.1. The identifications for $d = 1, 3$ follow from explicit substitution: at $d = 1$, $\chi_{-4}(1) = +1, \chi_{-4}(3) = -1$, and $w_K/4 = 1$, so $G_K = \Gamma(1/4)/\Gamma(3/4) = G^*$. At $d = 3$, $\chi_{-3}(1) = +1, \chi_{-3}(2) = -1$, and $w_K/4 = 3/2$, so $G_K = (\Gamma(1/3)/\Gamma(2/3))^{3/2} = R_3^{3/2}$. □

2. Structural features of the atlas

2.1. The $w_K = 2$ vs $w_K > 2$ dichotomy.

Observation. The two fields $\mathbb{Q}(i)$ ($d = 1$, $w_K = 4$) and $\mathbb{Q}(\rho)$ ($d = 3$, $w_K = 6$) are the only class-number-one imaginary quadratic fields with unit group strictly larger than $\{\pm 1\}$. They are therefore the only two members of the atlas for which the Chowla–Selberg exponent factor $w_K/4$ is non-trivially fractional (1 or 3/2, respectively), and the only two for which the underlying elliptic curve $E_K : y^2 = x^3 + (\text{shift})$ admits extra automorphisms.

Concretely, $\mathrm{End}(E_K) = \mathcal{O}_K$ in all nine cases, but $\mathrm{Aut}(E_K) = \mathcal{O}_K^\times$ has order 2, 4, or 6 depending on d. The corresponding squared automorphism counts $|\mathrm{Aut}|^2 \in \{4, 16, 36\}$ appear as the integer coefficients in the master quadratics of [1, §8] and [1, §15]; the seven $w_K = 2$ fields give the degenerate coefficient $|\mathrm{Aut}|^2 = 4$ and a correspondingly trivial master quadratic structure.

2.2. The Kronecker-character structure.

Theorem 2.1 (Character at each field)**.** *The Kronecker character χ_{d_K} of each class-number-one IQ field is:*

d	$\chi_{d_K} : (\mathbb{Z}/\lvert d_K\rvert\mathbb{Z})^\times \to \{\pm 1\}$
1	$\chi_{-4}(1) = +, \chi_{-4}(3) = -$
2	$\chi_{-8}(1) = +, \chi_{-8}(3) = +, \chi_{-8}(5) = -, \chi_{-8}(7) = -$
3	$\chi_{-3}(1) = +, \chi_{-3}(2) = -$
7	χ_{-7} : *quadratic residues mod 7 give* $+$; *non-residues* $-$
11	χ_{-11} *similarly via* $\left(\frac{a}{11}\right)$
19	χ_{-19} *via* $\left(\frac{a}{19}\right)$
43	χ_{-43} *via* $\left(\frac{a}{43}\right)$
67	χ_{-67} *via* $\left(\frac{a}{67}\right)$
163	χ_{-163} *via* $\left(\frac{a}{163}\right)$

For $d = 1, 3$: characters live on small finite groups of order ≤ 4, hence are exhausted by the lemniscatic and equianharmonic structures. For $d = 2$: $|d_K| = 8$ and the character has support of size 4 on $(\mathbb{Z}/8\mathbb{Z})^\times$. For $d \in \{7, 11, 19, 43, 67, 163\}$: $|d_K| = d$ is an odd prime, and χ_{d_K} is the Legendre symbol $\left(\frac{\cdot}{d}\right)$.

Proof. Standard. For d such that $-d \equiv 1 \pmod 4$ (i.e., $d \equiv 3 \pmod 4$), $d_K = -d$ and $\chi_{d_K}(a) = \left(\frac{a}{d}\right)$. For $d = 1, 2$ where $d \equiv 1, 2 \pmod 4$, $d_K = -4d$ and the character is given by the explicit formulas above. See [4, §11]. □

2.3. The j-invariants and Heegner numbers.

Observation. For each of the nine fields, the CM point τ_K (the unique CM representative in the fundamental domain of $\mathrm{SL}_2(\mathbb{Z})$ generating $\mathcal{O}_K$) has $j(\tau_K) \in \mathbb{Z}$. The nine integer j-values are

$$\{1728,\ 8000,\ 0,\ -3375,\ -32768,\ -884736,\ -884736000,\ -147197952000,\ -262537412640768000\}$$

corresponding to $d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$. The integers are factored as

d	Cubic-root factorisation of j
1	$j = 12^3$
2	$j = 20^3$
3	$j = 0^3$
7	$j = (-15)^3$
11	$j = (-32)^3$
19	$j = (-96)^3$
43	$j = (-960)^3$
67	$j = (-5280)^3$
163	$j = (-640320)^3$

The cubic-root structure $j(\tau_K) = m_K^3$ for $m_K \in \mathbb{Z}$ is a consequence of the $j = E_4^3/\Delta$ relation when Sym^2 of the class group is trivial; for $h_K = 1$ all nine cases yield cube integers.

2.4. **Heegner near-integers as manifestations of the atlas.**

Theorem 2.2 (Heegner near-integers explained)**.** *For each d in the atlas with $|d_K| \geq 7$, the quantity $e^{\pi\sqrt{d}}$ is exponentially close to $-j(\tau_K) + 744$:*

d	$e^{\pi\sqrt{d}} - (\lvert j\rvert + 744)$ *approximate*
7	$-47.07\ldots$
11	$-5.86\ldots$
19	$-0.22\ldots$
43	-2.23×10^{-4}
67	-1.31×10^{-6}
163	-7.50×10^{-13}

The accuracy improves rapidly with d because the next q-expansion coefficient of j is 196884 at q^1, and at $\tau_K = (1+\sqrt{-d})/2$ one has $|q| = e^{-\pi\sqrt{d}}$, so the correction is $\approx 196884\, q = 196884\, e^{-\pi\sqrt{d}}$. At $d = 163$ this is $196884 \cdot e^{-\pi\sqrt{163}} \approx 7.50 \times 10^{-13}$ — exactly the observed near-integer gap.

Proof. Direct application of the q-expansion $j(\tau) = q^{-1} + 744 + 196884\, q + 21493760\, q^2 + \ldots$ at $\tau = (1+\sqrt{-d})/2$, where $q = e^{2\pi i\tau} = -e^{-\pi\sqrt{d}}$ and hence $q^{-1} = -e^{\pi\sqrt{d}}$. Substituting:

$$j(\tau_K) = -e^{\pi\sqrt{d}} + 744 - 196884\, e^{-\pi\sqrt{d}} + O(e^{-2\pi\sqrt{d}}).$$

Since $j(\tau_K) = -|j(\tau_K)|$ for the seven fields with negative j-value, rearranging gives

$$e^{\pi\sqrt{d}} = |j(\tau_K)| + 744 - 196884\, e^{-\pi\sqrt{d}} + O(e^{-2\pi\sqrt{d}}).$$

The leading correction $-196884\, e^{-\pi\sqrt{d}}$ is the negative near-integer gap. $\square$

Remark. The Heegner near-integer for $d = 163$ is therefore not a numerical coincidence but a direct consequence of the $h_K = 1$ structure and the q-expansion of j. In atlas terms: the Chowla–Selberg constant $G_K(d = 163) = 1.13131\ldots$ is the "ratio-channel" companion of the "product-channel" near-integer $e^{\pi\sqrt{163}} \approx 640320^3 + 744$, both encoding the same χ_{-163}-character data.

3. The two distinguished fields and the dichotomy

3.1. **Lemniscatic ($d = 1$) and equianharmonic ($d = 3$).**

The dual-constant dichotomy of [1] — in which G^* (algebraic, ratio-channel) and $G_{\mathrm{G}} = 1/\mathrm{AGM}(1, \sqrt{2})$ (analytic, AGM-channel) are bridged by $G^* = 2\sqrt{\pi}\, G_{\mathrm{G}}$ — is specifically a feature of the $d = 1$ entry of the atlas. Its analog at $d = 3$ is the parallel structure

$$G_\rho \;=\; \frac{\Gamma(1/3)\,\Gamma(1/6)}{2\pi\sqrt{\pi}}, \qquad R_3 \;=\; \Gamma(1/3)/\Gamma(2/3),$$

linked by the equianharmonic Chowla–Selberg formula. The bridge identity in this case is $R_3 = \sqrt{3}\, G_\rho^2/\pi$ (a cubic analog of the lemniscatic bridge).

Theorem 3.1 (Dichotomy restricted to the atlas)**.** *A non-trivial dual-constant dichotomy — algebraic / ratio-channel constant linked to analytic / AGM-channel constant via a bridge identity $G_K = c_K \cdot \pi^{a_K} A_K^{b_K}$ for some analytic constant A_K — exists at exactly two members of the atlas: $d = 1$ (with $A_K = G_{\mathrm{G}}$) and $d = 3$ (with $A_K = G_\rho$). For the remaining seven fields ($d \in \{2, 7, 11, 19, 43, 67, 163\}$), G_K is the only canonical constant and there is no non-trivial AGM-style analytic counterpart at the same fundamental order.*

Sketch. The bridge identity of the lemniscatic and equianharmonic cases relies on the existence of an extra automorphism of E_K beyond ± 1, which furnishes a non-trivial endomorphism of the period lattice and induces the AGM-style iteration. For $w_K = 2$ the only automorphism is ± 1, the period lattice is rectangular without a non-trivial symmetry, and the AGM at the relevant modulus reduces to the classical (non-distinguished) case. The Chowla–Selberg constant G_K is still well-defined and canonical, but no analytic counterpart A_K is singled out. ☐

Remark. The seven $w_K = 2$ fields therefore have a *degenerate* version of the dichotomy of [1]: the algebraic-side G_K exists, but the analytic-side bridge counterpart does not (or, more precisely, collapses to a multiplicative constant times G_K itself). The dual constant story is genuinely *lemniscatic and equianharmonic only.*

3.2. **The atlas as a uniform statement.**

Theorem 3.2 (Atlas-uniform statement)**.** *For each K with $h_K = 1$, the Chowla–Selberg constant G_K and the Kronecker character χ_{d_K} jointly determine:*

(1) *the splitting law of rational primes in $\mathcal{O}_K$: a prime p splits in $\mathcal{O}_K$ iff $\chi_{d_K}(p) = +1$, is inert iff $\chi_{d_K}(p) = -1$, ramifies iff $\chi_{d_K}(p) = 0$;*
(2) *the Hecke L-function of the canonical CM elliptic curve $E_K/\mathbb{Q}$: its Euler product factors as a product over split-p Euler factors with non-trivial $a_p \neq 0$ and inert-p factors with trivial $a_p = 0$;*
(3) *the central value of the Dirichlet L-function: $L(\chi_{d_K}, 1)$ is a rational multiple of $\pi/\sqrt{|d_K|}$ (the analytic class-number formula at $h_K = 1$);*
(4) *the value G_K itself, via the χ-twisted Γ-product.*

All four data are projections of the single character χ_{d_K} at four distinct arithmetic levels (cf. [1, §16] for the χ_{-4} case).

Proof. (1) is class field theory of imaginary quadratic fields with $h_K = 1$; the splitting law of any prime is determined by the Kronecker symbol $\chi_{d_K}(p)$. (2) is the Hecke construction of $L(E_K/\mathbb{Q}, s)$ via the Hecke character ψ_{E_K} of K [5, §II]. (3) is the analytic class number formula $L(\chi_{d_K}, 1) = \pi h_K/(w_K\sqrt{|d_K|/4})$ for K imaginary quadratic. At $h_K = 1$ this gives explicit π-rational values $L(\chi_{-4}, 1) = \pi/4$, $L(\chi_{-3}, 1) = \pi/(3\sqrt{3})$, $L(\chi_{-8}, 1) = \pi/(2\sqrt{2})$, and so on. (4) is Definition 1.1. ☐

4. Closing remarks and open questions

The atlas extends the χ_{-4}-unification of [1] from the single lemniscatic case to all nine class-number-one imaginary quadratic fields. Each field contributes a canonical Chowla–Selberg constant G_K. Of these nine, two ($d = 1, 3$) have extra automorphisms and support the dual-constant dichotomy. The other seven contribute the constant G_K without an analytic counterpart at the same order, but participate fully in the Heegner phenomenon and in class field theory.

4.1. **Open questions specific to the atlas.**

A1. AGM analogs for the seven $w_K = 2$ fields. The bridge identity $G^* = 2\sqrt{\pi}\, G_{\mathrm{G}}$ exists at $d = 1$ because the lemniscatic elliptic curve admits an extra automorphism of order 4. At $d \in \{2, 7, 11, 19, 43, 67, 163\}$ the curve has only ± 1 automorphisms, and no AGM-style bridge is known. Determine whether any of the seven fields admits a non-AGM analytic identity that plays the bridge role.

A2. Master polynomials beyond the atlas. The master quadratic of [1, §8] has integer coefficient $|\mathrm{Aut}|^2 = 16$ at $d = 1$. For the seven $w_K = 2$ fields the analog coefficient is $|\mathrm{Aut}|^2 = 4$, yielding a trivial polynomial structure. Is there a non-trivial polynomial associated to each G_K in the $w_K = 2$ cases, perhaps via a higher Hecke operator or cyclotomic invariant?

A3. Joint L-value structure across the atlas. Each of the nine fields gives a critical L-value $L(\chi_{d_K}, 1) = c \cdot \pi/\sqrt{|d_K|}$. The constants c across the nine fields form a sequence of rationals; is there a uniform generating function for this sequence (e.g., a Dirichlet series over the atlas)?

A4. Heegner-style approximations for $d = 2, 7, 11$. The Heegner near-integers $e^{\pi\sqrt{d}}$ become arbitrarily close to an integer as d grows (asymptotically $\sim 196884 \cdot e^{-\pi\sqrt{d}}$). But for $d = 2, 7, 11$ the gap is order-10 to order-50, not "near-integer." What is the natural threshold (in d) at which the Heegner phenomenon becomes numerically striking?

References

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